PAN-ATLANTIC UNIVERSITY
SCHOOL OF SCIENCE AND TECHNOLOGY
Department of Computer Science

B.Sc. Software Engineering **Time allowed:** 45 Minutes
2025/2026 Academic Session (Second Semester Examination)
DTS304: Data Management I
Instructions: Answer **ALL Questions**

CASE STUDY: FleetTrack — Vehicle Rental Management System

FleetTrack is a vehicle rental company that operates across multiple city branches. The company needs a database to manage its fleet, customer records, rental agreements, and payment transactions. The following business rules define the system:

1. FleetTrack operates across multiple **BRANCHes**. Each BRANCH has a unique branch ID, a name, a city, an address, and a contact number.
2. Each **VEHICLE** is identified by a unique registration number and has a make, model, year of manufacture, and daily rental rate. A VEHICLE belongs to exactly one BRANCH, and a BRANCH can have many VEHICLEs assigned to it.
3. Every VEHICLE is classified under exactly one VEHICLE_CATEGORY (e.g., Economy, SUV, Luxury, Van). A **VEHICLE_CATEGORY** is identified by a category ID and has a category name. A single VEHICLE_CATEGORY may classify many VEHICLEs.
4. **CUSTOMERs** register with the company. Each CUSTOMER is identified by a unique customer ID and has a full name (stored as first name and last name), driver's licence number, phone number, and email address.
5. A CUSTOMER may enter into many RENTAL_AGREEMENTs over time. Each **RENTAL_AGREEMENT** involves exactly one CUSTOMER and exactly one VEHICLE. A RENTAL_AGREEMENT has a unique agreement ID and records a start date, expected return date, an actual return date (which may be null if the vehicle has not yet been returned), and a rental status (Active, Returned, or Overdue).
6. **PAYMENTs** are recorded against RENTAL_AGREEMENTs. PAYMENT cannot exist without the RENTAL_AGREEMENT it belongs to — it is therefore a weak entity. Each PAYMENT has a payment number (partial key), a payment date, an amount, and a payment method (e.g., Cash, Card, Bank Transfer).

Question 1 [15 marks]

a. Using the business rules in the case study, draw a complete **ER diagram** for the FleetTrack system. Your **diagram** must clearly show:

   - All entities and their attributes, with primary keys correctly identified. Identify and represent the attributes accordingly in your ER Diagram based on your interpretation.

- All relationships between entities, with correct cardinality constraints.

[9 marks]

b. Convert your ER diagram from **Question 1(a)** into a complete relational database schema. Present your answer as a set of table definitions in the format below:

```
Table1(Attribute1, Attribute2, Attribute3, …)
Table2(Attribute1, Attribute2:Table1, Attribute3, …)
```

In the format above, ***Attribute2*** in ***Table2*** is a foreign key, referencing ***Table1***.

[6 marks]

Question 2 [10 marks]

a. The lead developer of ***FleetTrack*** proposes using MongoDB instead of a relational DBMS, arguing that document databases are more flexible.

i. Recommend **FOUR (4)** Document DBMS that could be used. **[2 marks]**

ii. Explain TWO reasons why a relational database would be more appropriate for the ***FleetTrack*** system described in this case study.

[2 marks]

b. Use **your own** schema in **Question 1(b)** to answer the following SQL questions:

i. Write the **SQL CREATE TABLE** statement for the **RentalAgreement** table based on your knowledge of Primary key, Foreign key, and other relevant associated constraints learned in class. **[3 marks]**

ii. Write a **SELECT** query to retrieve the registration number, make, model, and daily rate of all vehicles assigned to the branch with BranchID 'BR003', sorted by daily rate in **ascending order**.

[1.5 marks]

iii. The vehicle with registration number 'ABJ-221-GH' has been relocated to the branch with BranchID 'BR007'. Write an **UPDATE** statement to reflect this change in the Vehicle table. **[1.5 marks]**